

Standard mode

$\text{refl}(R, A, B) :- \text{prodcart}(A, B, P)$
 $\text{sublista}(R, P)$
 $\text{rellinara}(R, A) :- \text{rellin}(R, A, A)$
 $\text{diag}(A, D) :- \text{setof}((X, X), \text{member}(X, A), D)$
 $\text{diagonala}([], []) .$
 $\text{diagonala}([H|T], [(H, H)|L]) :- \text{diagonala}(T, L)$

$\text{refl}(R, A) :- \text{not}(\text{member}(X, A), \text{not}(\text{member}((X, X), R)))$
 $\text{reflexiva}(-, []) .$
 $\text{reflexiva}(R, [H|T]) :- \text{member}((H, H), R), \text{reflexiva}(R, T)$

$\text{sim}(R) :- \text{not}(\text{member}((X, Y), R), \text{not}(\text{member}((Y, X), R)))$
 $\text{simetra}(R) :- \text{auxsim}(R, R)$
 $\text{auxsim}(-, []) .$
 $\text{auxsim}(R, [(X, Y)|T]) :- \text{member}((Y, X), R), \text{auxsim}(R, T)$

$\text{trans}(R) :- \text{not}(\text{member}((X, Y), R), \text{member}((Y, Z), R), \text{not}(\text{member}(X, Z), R))$

$R \subseteq A \times B \Leftrightarrow R = S \Leftrightarrow R^{-1} = S^{-1}$
 $R \subseteq S \Leftrightarrow (R \rightarrow R^{-1}, S \rightarrow S^{-1})$
 $R \subseteq A \times B \Leftrightarrow (R^{-1})^{-1} = R$
 $R \subseteq B \times A$

$(a, b) \in R \Leftrightarrow (b, a) \in R^{-1}$
 $R^{-1} \subseteq S \Leftrightarrow (a, b) \in R \Rightarrow (b, a) \in S$
 $(a, b) \in R^{-1} \Leftrightarrow (b, a) \in R$

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$R \subseteq A \times A = A^2 \Leftrightarrow R^{-1} \subseteq A^2$
 $R \rightarrow \text{refl} \Leftrightarrow \Delta_A \subseteq R$

$R \rightarrow \text{sim} \Leftrightarrow (\forall x, y \in A) ((x, y) \in R \Rightarrow (y, x) \in R) \Leftrightarrow R \subseteq R^{-1} \Leftrightarrow R^{-1} \subseteq (R^{-1})^{-1} = R \Leftrightarrow R \subseteq R^{-1} \subseteq R$
 $(p \Leftrightarrow q) \Rightarrow (p \Rightarrow q \Leftrightarrow (p \wedge q))$
 $R = R^{-1}$

$R \rightarrow \text{trans} \Leftrightarrow (\forall x, y, z \in A) ((x, y) \in R \wedge (y, z) \in R \Rightarrow (x, z) \in R) \Leftrightarrow R \circ R \subseteq R$
 $\text{Fie } x, y, z \in A$

$\xRightarrow{h} \text{ De } (x, y) \in R \circ R, \Leftrightarrow (\exists u \in A) (x, u) \in R \wedge (u, y) \in R$
 $\xRightarrow{h} \text{ De } (x, y), (y, z) \in R \Rightarrow (x, z) \in R \circ R \xRightarrow{(R \circ R) \subseteq R} (x, z) \in R$

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$R \subseteq A \times B, S \subseteq B \times C \Rightarrow S \circ R \subseteq A \times C$
 $A=B=C, R, S \subseteq A \times A$
 $comp(S, R, S \circ R) :- setof((X, Z),$
 $X \wedge (member(X, Y), R, member(Y, Z), S), S \circ R), !.$
 $comp(, , []).$
 $prodrel(R, A, R \times A) :- setof(((X1, X2), (Y1, Y2)),$
 $(member(X1, Y1), R, member(X2, Y2), A), R \times A), !.$
 $prodrel(, , []).$
 $putere(R, 1, R). % putere(, A, 0, D) :- diag(A, D).$
 $putere(R, N, RelN) :- N > 1, PN is N-1, putere(R, PN, RelPN), comp(RelPN, R, RelN),$
 $putere(R, N, RelN) :- N < 0, K is -N, invrel(R, I), putere(I, K, RelN).$

Diagram illustrating the composition of relations R and S to form $S \circ R$. The diagram shows sets A, B, C and relations R, S as directed graphs. The composition $S \circ R$ is shown as a new relation.

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$preord(R, A) :- refl(R, A), tranz(R).$
 $eg(R, A) :- preord(R, A), sur(R).$
 $releg(R, A) :- rellinara(R, A), eg(R, A).$
 $relatieg(A, EgA) :- setof(R, releg(R, A), EgA),$
 $?- relatieg([g1, g2], L), oflistat(L), length(L, N).$
 $parteg(E, A, P) :- setof(C, X \wedge (member(X, A),$
 $clasa(X, E, C)), P),$
 $clasa(X, E, C) :- setof(Y, member(X, Y), E, C).$
 $partitu(A, PartA) :- setof(P, (releg(R, A), parteg(R, A, P)),$
 $PartA),$
 $?- partitu([g1, g2], L), oflistat(L), length(L, N).$

Diagram illustrating the partitioning of a set A into equivalence classes. The diagram shows a set A and its partitioning into equivalence classes EgA . The partitioning is done using the relation $releg$ and the function $parteg$.

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$A := Eg_{g1, g2}$
 $|A| = 4$
 $(A_i)_{i \in I} \in Part(A)$
 $\downarrow 2$
 $UA_i \in Eg(A)$

Diagram illustrating the partitioning of a set A into equivalence classes. The diagram shows a set A and its partitioning into equivalence classes EgA . The partitioning is done using the relation $releg$ and the function $parteg$.

eg part $([], []) ,$

eg part $([A | LA], E) :- \text{predcat}(A, A, A, A)$

eg part $(LA, F), \text{reunite}(A, A, F, E)$

antisym $(R) :- \text{not}((\text{member}(X, Y), R)$

$\text{member}(X, Y), X \neq Y))$

asim $(R) :- \text{not}((\text{member}(X, Y), R)$
 $\text{member}(Y, X), R))$

$\text{isrefl}(R) :- \text{not}(\text{member}((X, X), R))$

$\text{ord}(R, A) :- \text{presd}(R, A), \text{antisym}(R)$

$\text{reld}(R, A) :- \text{rellnva}(R, A), \text{ord}(R, A)$

$\text{ordstr}(R) :- \text{asim}(R), \text{trans}(R)$

$\text{ordstricta}(R) :- \text{isrefl}(R), \text{trans}(R)$

$\leq = \leq \Delta_A$
 $\text{Ord}(A) := \text{mult rel de ord pe mult A}$

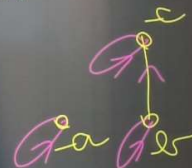
$(\forall a, b \in A) (a < b \Leftrightarrow \begin{cases} a < b \\ a = b \end{cases})$

$\leq = \leq \cup \Delta_A$
 $(\forall a, b \in A) (a \leq b \Leftrightarrow \begin{cases} a < b \\ a = b \end{cases})$

$\text{ordstrdnord}(S, \emptyset) :- \text{setof}((X, Y), (\text{member}(X, Y), S), !)$

$\text{ordstrdnord}([], \rightarrow, X \neq Y), S, !$

$\text{orddnordstr}(\emptyset, S, A) :- \text{diag}(A, D), \text{reunite}(S, D, \emptyset)$



$\rightarrow = \{(a, b) \in A \times A \mid (\exists x \in A) (a < x < b)\}$

Van reduct: dacă $|A| < \aleph_0$, atunci:

$\leq = \mathcal{T}(\rightarrow), \leq = \mathcal{R}(\leq) = \mathcal{R}(\mathcal{T}(\rightarrow)) = \text{Pre}(\rightarrow)$

$\text{succdnordstr}(\text{Succ}, S) :- \text{setof}((X, Y)$

$\text{not}(\text{member}(X, U), S), \text{member}(U, Y), S), \text{Succ}, !$

$\text{succdnordstr}([], \rightarrow)$

$\text{succdnord}(\text{Succ}, \emptyset) :- \text{setof}((X, Y)$

$\text{not}(\text{member}(X, U), \emptyset), X \neq U$

$\text{member}(U, Y), \emptyset), U \neq Y), \text{Succ}, !$

$\text{succdnord}([], \rightarrow)$

$\text{relatiord}(A, \text{Ord} A) :- \text{setof}(R, \text{reld}(R, A), \text{Ord} A)$

$? - \text{relatiord}([a, b, c], L), \text{ofisista}(L), \text{length}(L, N)$